

# ALGORITHM

Autumn 2026, Quiz 7, April 29

11:45-12:05 am

9

姓名: 黃翊宏  
學號: 01357101  
系所: CS2B

1. Please answer the following questions:

- Using the figure in page 12 of Lecture Note 8 as a model, illustrate the operation of COUNTING-SORT on the array  $A = \langle 5, 0, 2, 2, 1, 3, 7, 6, 4, 8, 2 \rangle$ .
- Using the figure in page 15 of Lecture Note 8 as a model, illustrate the operation of RADIX-SORT on the following numbers: 257, 846, 683, 462, 268, 674, 358, 921, 331, 536.
- Using the figure in page 21 of Lecture Note 8 as a model, illustrate the operation of BUCKET-SORT on the array  $A = \langle .79, .13, .16, .64, .39, .20, .89, .53, .71, .42 \rangle$ .

(a) 寫在背面

(b)

| ORIG    | ↓         | ↓         | ↓    |
|---------|-----------|-----------|------|
| 257 ✓   | 9 2 1 ✓   | 9 2 1     | 2 57 |
| 846 ✓   | 3 3 1 ✓   | 3 3 1 ✓   | 2 68 |
| 683 ✓   | 4 6 2 ✓   | 5 3 6 ✓   | 3 31 |
| 462 ✓   | 6 8 3 ✓   | 8 4 6 ✓   | 3 58 |
| 268 ✓ → | 6 7 4 ✓ → | 2 5 7 ✓ → | 4 62 |
| 674 ✓   | 8 4 6 ✓   | 3 5 8 ✓   | 5 36 |
| 358 ✓   | 5 3 6 ✓   | 4 6 2 ✓   | 6 74 |
| 921 ✓   | 2 5 7 ✓   | 2 6 8 ✓   | 6 83 |
| 331 ✓   | 2 6 8 ✓   | 6 7 4 ✓   | 8 46 |
| 536 ✓   | 3 5 8 ✓   | 6 8 3 ✓   | 9 21 |

(c)

| A     | B               | sort B | B               |
|-------|-----------------|--------|-----------------|
| .79 ✓ | 0 /             |        | 0 /             |
| .13 ✓ | 1 0 → .13 → .16 |        | 1 0 → .13 → .16 |
| .16 ✓ | 2 0 → .20       |        | 2 0 → .20       |
| .64 ✓ | 3 0 → .39       |        | 3 0 → .39       |
| .39 ✓ | 4 0 → .42       |        | 4 0 → .42       |
| .20 ✓ | 5 0 → .53       |        | 5 0 → .53       |
| .89 ✓ | 6 0 → .64       |        | 6 0 → .64       |
| .53 ✓ | 7 0 → .79 → .71 |        | 7 0 → .71 → .79 |
| .71 ✓ | 8 0 → .89       |        | 8 0 → .89       |
| .42 ✓ | 9 /             |        | 9 /             |

按順序放  $\langle .13, .16, .20, .39, .42, .53, .64, .71, .79, .89 \rangle$

(a) Counting Sort < 5, 0, 2, 2, 1, 3, 7, 6, 4, 8, 2 >

~~5~~ ~~0~~ ~~2~~ ~~2~~ ~~1~~ ~~3~~ ~~7~~ ~~6~~ ~~4~~ ~~8~~ ~~2~~  
A < 5, 0, 2, 2, 1, 3, 7, 6, 4, 8, 2 >

C<sup>0 1 2 3 4 5 6 7 8</sup>  
< 1, 1, 3, 1, 1, 1, 1, 1, 1 > → 計算元素有幾個

↳ C<sup>0 1 2 3 4 5 6 7 8</sup>  
< 1, 2, 5, 6, 7, 8, 9, 10, 11 > → ≤ i 的有幾個 (前綴)

原陣列 A 由右往左掃一次  
將 A[idx] 的值視為 C  
的 idx, 並將結果視為 B  
的 idx 將元素放入。

B 1 2 3 4 5 6 7 8 9 10 11  
2

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 2 ~~5~~ 6 7 8 9 10 11  
4

↓

B 1 2 3 4 5 6 7 8 9 10 11  
2 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 2 4 6 7 8 9 10 ~~11~~  
10

↓

B 1 2 3 4 5 6 7 8 9 10 11  
2 4 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 2 4 6 ~~7~~ 8 9 10 10  
6

↓

B 1 2 3 4 5 6 7 8 9 10 11  
2 4 6 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 2 4 6 6 8 ~~9~~ 10 10  
8

↓

B 1 2 3 4 5 6 7 8 9 10 11  
2 4 6 7 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 2 4 6 6 8 8 ~~10~~  
9

↓

B 1 2 3 4 5 6 7 8 9 10 11  
2 3 4 6 7 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 2 4 ~~6~~ 6 8 8 9 10  
5

↓

B 1 2 3 4 5 6 7 8 9 10 11  
1 2 3 4 6 7 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 ~~2~~ 4 5 6 8 8 9 10  
1

↓

B 1 2 3 4 5 6 7 8 9 10 11  
1 2 2 3 4 6 7 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 1 ~~4~~ 5 6 8 8 9 10  
3

↓

B 1 2 3 4 5 6 7 8 9 10 11  
1 2 2 2 3 4 6 7 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
1 1 ~~3~~ 5 6 8 8 9 10  
2

↓

B 1 2 3 4 5 6 7 8 9 10 11  
0 1 2 2 2 3 4 6 7 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
~~1~~ 1 2 5 6 8 8 9 10  
0

↓

B 1 2 3 4 5 6 7 8 9 10 11  
0 1 2 2 2 3 4 5 6 7 8

C<sup>0 1 2 3 4 5 6 7 8</sup>  
0 1 2 5 6 8 ~~9~~ 10 X -1  
7

↓

B  
→ < 0, 1, 2, 2, 2, 3, 4, 5, 6, 7, 8 >